

Zero-shot time series foundation models for statistical postprocessing of weather forecasts: how much training data do they replace?

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Abstract

Statistical postprocessing of ensemble forecasts is usually trained on multi-year, per-station archives; a zero-shot time series foundation model needs none. Using EUPPBench v1.0’s German 2m temperature subset (49 stations, 0–120 h; all methods scored on 737,809 matched instances), we compare a frozen TimesFM-3 (ensemble mean as a past-future covariate) against Ensemble Model Output Statistics (EMOS) and a one-parameter variance-inflation rescaling. At full training data, every trained method beats TimesFM-3 on the Continuous Ranked Probability Score (CRPS); so does a fixed-climatological rescaling using no training data at all. Pooled EMOS’s CRPS crosses TimesFM-3’s at 1.6 cases (6 days), though the crossing is significant only from 3 cases (10 days) onward, for both EMOS variants, under day-blocking. 80%-nominal coverage crosses more slowly (5.1x and 32.0x later than CRPS, pooled and local EMOS). TimesFM-3 retains a calibration-sharpness advantage: even a leakage-optimistic multiplier fit on test-set coverage needs 0.3965 K more at matched coverage (95% CI [0.2887, 0.4997]). Its short-lead CRPS advantage reverses by 43.49 h; a persistence-augmented EMOS closes only 3.4% of the gap, which does not support an information-asymmetry explanation, though not from architecture. Supplying ensemble spread as a second covariate does not close the calibration gap either: it worsens CRPS and reaches nominal coverage only with a wider interval. Overall, on CRPS this frozen model is mostly not competitive with a few days of station-specific training data – with one exception that survives even a leakage-optimistic bound, and a short-lead advantage that the information-asymmetry explanation does not account for.

1 Introduction

Numerical weather prediction (NWP) output is systematically biased at any single station or grid point: model resolution, physical parameterizations, and terrain representation err in a station-specific and often persistent way, and statistical postprocessing exists specifically to correct this systematic component rather than relying on raw ensemble output directly [Vannitsem et al., 2021, Gneiting et al., 2005]. Beyond Ensemble Model Output Statistics (EMOS)’s original Gaussian regression [Gneiting et al., 2005], the field has extended postprocessing along several axes: mixture and non-Gaussian predictive families [Baran and Lerch, 2016], target-specific reformulations for variables EMOS’s Gaussian assumption fits poorly [Scheuerer, 2013], explicit calibration-enforcing extensions [Wilks, 2017], and permutation-invariant neural architectures that treat ensemble members as an unordered set [Höhlein et al., 2024]. The Distributional Regression Network (DRN, Rasp and Lerch, 2018) is this paper’s own neural baseline (Section 3).

Every one of these methods shares the same requirement: a multi-year archive of paired ensemble/observation history at the specific station being corrected. Vannitsem et al. [2021]’s

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review of the field flags this as a structural limitation, not an incidental one – and it is exactly what a newly commissioned station, a station following an NWP model version upgrade that breaks the archive’s stationarity, or a station in a sparsely observed region cannot supply. Trained postprocessing’s accuracy gains, in other words, come with an entry cost that a purely zero-shot alternative would not.

Time series foundation models (TSFMs) – large, pretrained sequence models applied without any task-specific training or parameter updates on the target series – offer a structurally different answer: no station-specific training archive at all. This is a different proposition from the spatial, full-field AI weather-prediction models that have separately transformed NWP itself [Lam et al., 2023, Price et al., 2024, Bodnar et al., 2025]: those models replace the numerical forecast; a TSFM used for postprocessing takes an existing ensemble forecast as input and corrects it at the station level, one series at a time, with no spatial field anywhere in its architecture. TimesFM-3 – the 3.0 release of Das et al.’s decoder-only TimesFM architecture [Das et al., 2024] – is one such model, and its past-future covariate support (a version-specific capability sourced directly in Section 3.2, not part of the original architecture paper) is the specific capability this paper exploits: the ensemble forecast can be supplied as a covariate known across the entire forecast horizon, not just in the past, letting a zero-shot model use NWP guidance the way a postprocessing method would, without training on this benchmark’s data at all.

To our knowledge, no existing work directly measures the training-data size at which a station’s own trained postprocessing model overtakes a zero-shot time series foundation model given the same NWP guidance – the question this paper answers. On the identical EUPPBench benchmark this paper uses, Van Poecke et al. [2025], Mlakar et al. [2024], and Landry et al. [2025] each propose a trained neural or flow-based postprocessing architecture, and Allen et al. [2024] evaluates multivariate calibration on it – none compares against a zero-shot foundation model. Baran and Mihalina [2026] studies the same sharpness axis this paper’s calibration-sharpness result depends on, on the same benchmark, but not the zero-shot-vs-trained breakpoint question. Sun and Sun [2026] pose a structurally identical question – is a zero-shot time series foundation model competitive with domain-trained methods – in hydrology, not weather postprocessing; Bharadwaj et al. [2026] benchmark TSFM zero-shot skill across regions in air quality forecasting, a related but distinct target variable and evaluation design. Wessel et al. [2024] and Muschinski et al. [2023] are the closest small-training-data postprocessing rivals to this paper’s own breakpoint curve, targeting the same low-data regime with different (lead-time-continuous, random-forest) trained architectures rather than a zero-shot comparison. None of these measures where the crossing point actually is.

This paper’s contribution is threefold. (1) At full training data, every trained method (EMOS pooled, EMOS local, DRN) beats TimesFM-3’s Continuous Ranked Probability Score (CRPS, Section 4.1). A one-parameter variance-inflation rescaling of the raw ensemble does too: a fixed-climatological variant beats it with no training data at all, and the CRPS-optimal variant beats it at every training size we tested it at, from 30 days upward (Section 4.1) – on CRPS, the answer to this paper’s central question is negative for TimesFM-3, and not narrowly so. This advantage starts at very small training sizes: pooled EMOS’s CRPS crosses TimesFM-3’s at 1.6 cases (6 calendar days, Section 4.2), but that raw crossing is not itself statistically significant; the advantage is significant from 3 cases (10 calendar days) onward under day-blocking, our primary block definition, for both EMOS pooled and EMOS local alike – the station-blocked t -test does not reach significance anywhere in the tested $k = 1$ –9 range, while the station-blocked Wilcoxon test does from $k = 2$ onward, the same test-statistic divergence already reported at 9 cases (Section 4.5), now shown to persist across the whole low- k range. Coverage crosses more slowly (ratios computed from the unrounded breakpoint estimates, not the values as displayed): pooled EMOS’s coverage crosses TimesFM-3’s roughly 5.1x later than its CRPS does (8.4 cases / 29 days), and local EMOS’s about 32.0x later (56.2 cases / 196 days) – so "a few days is enough" holds for CRPS specifically, not for calibration. These comparisons are between methods given different infor-

mation: TimesFM-3’s covariate channel received the ensemble mean but not ensemble spread in this study, while EMOS and DRN received spread but no observation history (Section 3.2, Table 1); supplying spread as a second covariate does not close this calibration-sharpness gap either (Section 5.4). (2) On the calibration-sharpness axis, TimesFM-3 holds a real advantage over a trivial rescaling: even a leakage-optimistic multiplier fit directly on test-set coverage – an upper bound no legitimately deployable trivial rescaling could exceed – needs 5.5344 K, still wider than TimesFM-3’s 5.1379 K (Section 5.2). (3) TimesFM-3’s short-lead CRPS advantage first flips sign at 22.36 h and reverses durably by 43.49 h (Section 4.3); the mechanism behind that advantage has two independent parts. The observation-history part was tested directly and does not support an information-asymmetry explanation – a persistence-augmented EMOS closes only 3.4% of the gap – but the test itself cannot cleanly distinguish an information-set explanation from an architectural one (Section 5.3). The ensemble-spread/calibration part was also tested directly, by supplying TimesFM-3 with ensemble spread as a second covariate: it worsens CRPS and shifts the model along the calibration-sharpness trade-off without closing the calibration deficit, even though the channel demonstrably accepts and uses the input (Section 5.4). Together these results argue that a frozen, zero-shot time series foundation model is, for this task, mostly not competitive with a few days of station-specific training data on CRPS – with one genuine exception that survives even a leakage-optimistic trivial baseline (the calibration-sharpness axis), and a short-lead advantage that the information-asymmetry explanation does not account for. The evidence presented here is for one time series foundation model, TimesFM-3; whether it generalizes to other pretrained time series models (e.g. Chronos-2, Moirai-2) is untested.

2 Data and experimental design

We use the EUPPBench postprocessing benchmark, version v1.0 [Demaeyer et al., 2023], restricted to its German subset of 49 stations and the 2m temperature target. EUPPBench defines its own train/test split, which we never re-partition: training uses ECMWF reforecasts issued 1997–2016 with 11 ensemble members, and testing uses operational ECMWF forecasts issued 2017–2018 with 51 members. Both sides carry 21 lead times from 0 to 120 h in 6-hour steps. Reference-quality observations are DWD station measurements; the ensemble side is ECMWF’s operational and reforecast archives as distributed through EUPPBench. EUPPBench also defines a wind-speed target (w_{10}); we defer it here because no country in the benchmark provides a direct wind-speed observation series, only wind gust, and substituting gust for speed would change what is being verified.

TimesFM-3’s fixed context-window requirement (at least 40 prior observations) excludes some station/lead/issue-time combinations that would otherwise be evaluable by the other methods, so every comparison in this paper is computed on the intersection of instances all methods can produce a prediction for, not each method’s own largest possible set. The full test set otherwise available to EMOS and the raw ensemble contains 748,099 instances; requiring TimesFM-3’s context window reduces this to 737,809 matched instances, and every result in this paper is computed on that matched set for every method, including EMOS and the raw ensemble, not their own larger native set.

The archive’s raw station dimension carries 51 station identifiers; two are excluded throughout this study because their observation series contain missing values that would otherwise propagate NaNs into every downstream metric, leaving the 49 stations used everywhere in this paper. Ensemble summary statistics (mean and variance) are computed once from the raw per-member forecasts before any postprocessing method sees them, using an unbiased ($ddof = 1$) variance estimator on both the training and test sides so that EMOS is never trained against a systematically different variance definition than the one it is evaluated against. The training-side ensemble variance is nonetheless estimated from 11 members against 51 on the test side; an 11-member variance estimate is intrinsically noisier than a 51-member one of the same true

spread, so EMOS is trained against a noisier estimate of ensemble variance than the one used to evaluate it (Section 5.7).

The paper’s central axis is training-data size, expressed as a number of training *cases* k rather than calendar days directly, since the reforecast archive samples a fixed, non-daily grid of issue dates rather than every calendar day. The reforecast archive’s raw `time` coordinate cannot be reliably decoded to calendar dates: attempting to decode it raises an `OverflowError`, confirmed to be caused by an unmasked fill-value sentinel surviving into the time-unit conversion, not by a transient network fault, since the same failure text was independently confirmed on the sibling reforecast-observations file. We therefore treat each issue date as an abstract $(\text{year_idx}, \text{time_idx})$ pair rather than a decoded calendar date: 209 representative issue-date positions (time_idx) recur once per analog year (year_idx , 20 analog years in total), so the full archive contains 4180 cases (209×20). We measured the calendar-day span of one such 209-date cycle directly from its real, decodable date labels: it spans 728 calendar days – consistent with a twice-weekly issue cadence over two calendar years, not one – giving 3.4833 calendar days per case. We convert a requested training size of N days to $k = \text{round}(N/3.4833)$ cases throughout, and report both units together wherever a training size is cited, since case count and calendar-day equivalent are not interchangeable at this archive’s actual sampling density. The contiguous training arm (below) orders cases by descending $(\text{year_idx}, \text{time_idx})$, assuming ascending year_idx is chronologically more recent; only time_idx ’s ordering was independently verified against real decoded dates, so this assumption is disclosed, not confirmed (Section 5.7 bounds its limited practical impact).

We test two ways of selecting which k cases make up a given training size, since they answer different questions. The *contiguous* arm (primary) takes the k cases closest in time to the test period, working backward – the realistic scenario for a newly deployed station whose limited history is chronologically contiguous, so small k sees substantially less seasonal variety than a full archive would. The *random* arm (secondary, 5 seeds, mean and standard deviation reported) draws k cases uniformly at random from the full archive, giving artificial full-season coverage even at small k . The gap between the two arms’ results is itself a reported finding about the operational value of seasonal training coverage (Section 4), not a discrepancy to be resolved in favor of one arm. Baran and Mihalina [2026] targets the same sharpness axis on the same benchmark; we discuss the relationship to our own sharpness results directly in Section 5 rather than here. Other EUPPBench applications are surveyed in Section 1.

3 Methods

3.1 Postprocessing methods compared

We compare six families of methods, all sharing a single interface: given a station’s ensemble forecast summary for a lead time, produce a predictive distribution for the observed 2 m temperature at that lead, expressed as the 9 quantiles 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 (a fixed global constant used identically by every method in this study).

Raw ensemble. The empirical distribution of the raw NWP ensemble members at each station and lead time, used unmodified. This is the zero-shot, zero-parameter baseline every postprocessing method is expected to improve on.

EMOS. Ensemble Model Output Statistics [Gneiting et al., 2005], a nonhomogeneous Gaussian regression fit by minimizing the closed-form Gaussian CRPS [Gneiting and Raftery, 2007] over the training set: $\mu = a + b \cdot \bar{x}$, $\sigma^2 = \exp(c) + \exp(d) \cdot s^2$, where $\bar{x}$ and s^2 are the ensemble mean and variance. We fit two variants of this same model, differing only in how many stations share one parameter set: *pooled* EMOS fits one global (a, b, c, d) across all 49 stations, following the field’s standard practice [Gneiting et al., 2005, Raftery et al., 2005]; *local* EMOS fits a separate (a, b, c, d) per station whenever that station has enough training rows to do so (we

require at least 5 rows; stations below this threshold are excluded from local EMOS’s evaluation at that training size, and the resulting station-coverage fraction is reported alongside every local-EMOS result). Baran and Lerch [2018] and Gebetsberger et al. [2018] report comparable station-adaptive EMOS variants at small training sizes, the regime this paper’s breakpoint curve targets directly.

DRN. The Distributional Regression Network of Rasp and Lerch [2018]: a feed-forward neural network with a learned per-station embedding, taking the ensemble mean and variance as input and outputting $(\mu, \log \sigma^2)$ parameters of the same Gaussian family EMOS uses, trained by gradient descent (Adam) on the same closed-form Gaussian CRPS loss, reimplemented in PyTorch so it is differentiable. Unlike the original DRN, which additionally uses auxiliary NWP fields and lead time as inputs, our training pipeline pools all 21 lead times into one fit and does not expose lead time to any model (Section 5.7); DRN and EMOS are therefore compared under the same feature constraint, not DRN’s originally-intended full feature set. Schulz and Lerch [2022] runs a systematic comparison of neural postprocessing architectures under this same feature-constrained setup, the closest methodological precedent to our own DRN evaluation. Time-series/lead-time-continuous EMOS variants [Jobst et al., 2024, Wessel et al., 2024] target the small-training-data and lead-time-resolved regimes this paper’s breakpoint curve also explores, without themselves being baselines here (a wider survey of the neural-postprocessing and quantile-regression literature is given in Section 1).

TimesFM-3 (zero-shot). The 3.0 release of TimesFM, a frozen, pretrained decoder-only time series foundation model. The architecture family is described in Das et al. [2024]; that paper describes the original TimesFM 1.0 decoder-only architecture and does not cover the 3.0 release’s covariate support, which Section 3.2 sources directly rather than through that citation. This project uses checkpoint `google/timesfm-3.0-pytorch`, with no fine-tuning, task-specific training, or parameter updates of any kind. Section 3.2 describes its input construction. Contrasted with generalist AI weather-prediction models that operate on full spatial fields [Lam et al., 2023, Price et al., 2024, Bodnar et al., 2025] – TimesFM-3 is used here in univariate, single-station mode, not as a spatial forecasting system, and is not compared against those models here.

Variance-inflation baseline. A trivial, zero/low-parameter alternative: scale the raw ensemble’s per-instance spread by a single multiplier λ and read off Gaussian quantiles at the same 9 levels. We report 4 variants, each answering a different question about whether TimesFM-3’s apparent skill is trivially reproducible [Zhang and Gilpin, 2025]: (a) a CRPS-optimal λ fit by minimizing training-set CRPS at each training size N (fair comparison against EMOS, which also uses N training cases); (b) a fixed climatological λ , requiring no data at all (fair comparison against TimesFM-3’s own zero-shot status); (c) a coverage-matched λ , fit on training data only (never on test-set coverage) by root-finding to reach as closely as a training-fit multiplier allows TimesFM-3’s own empirical coverage, then evaluated for interval width on the test set – this directly asks whether TimesFM-3’s calibration, not just its CRPS, is achievable by a one-parameter rescaling. Because λ is fit on training data and applied to a different test set, the coverage this variant actually reaches on test does not land exactly on the training-time target; we report both values together (Section 4.1) rather than the word “exactly.” (d) a leakage-optimistic λ , fit by root-finding directly on *test-set* coverage (never on training data) to reach TimesFM-3’s own empirical coverage – the deliberately leakage-contaminated counterpart to (c), answering not "what can a legitimate, leakage-free trivial rescaling achieve" but "what is the ceiling on what any trivial rescaling could ever achieve at this coverage target, even with an unfair advantage." Variant (d) is never presented as a legitimate baseline result; it exists solely to bound how much of variant (c)’s width disadvantage against TimesFM-3 (Section 4.1) could in principle be a leakage artifact of the train/test split rather than a real limit of trivial rescaling.

3.2 TimesFM-3 covariate construction, and the information asymmetry it creates

TimesFM-3’s target series is a station’s own observation history; its past-future (dynamic) covariate channel¹ is filled with the ensemble mean, supplied as a value known across the entire forecast horizon rather than only in the past. This study supplied only the ensemble mean to this channel; ensemble spread was not. This is an input-design choice, not a confirmed architectural limit: the installed package’s own source documents the channel’s accepted shape as genuinely multi-channel, and a preliminary test confirms the model’s output is measurably sensitive to a second channel’s real informational content, not merely its presence (Section 5.7 reports this preliminary test; Section 5.4 reports the full evaluation it motivated). The model is entirely frozen: no fine-tuning, no gradient updates, and no task-specific training of any kind are performed at any point in this study. Kreusel et al. [2026] independently reports that covariate handling is the dominant factor in time series foundation model performance; a coding error encountered during this project’s own development, where an incoherent covariate channel measurably degraded TimesFM-3’s apparent skill before being corrected, is consistent with that finding (Section 5.5).

This covariate construction leaves TimesFM-3 and the trained statistical methods (EMOS, DRN) with genuinely different information, as this study was actually run: TimesFM-3 sees the station’s own recent observation history but was not given ensemble spread; EMOS and DRN see ensemble spread but no observation history at all (Table 1). We treat this asymmetry as a candidate explanation for three separate empirical results in this paper (Section 4): TimesFM-3’s short-lead advantage (recent observations carry strong persistence information at short lead times, information EMOS/DRN were not given observation history to use in this study’s configuration), its calibration deficit relative to EMOS (TimesFM-3’s covariate channel was not given ensemble spread, the direct measure of forecast uncertainty, in this study), and its long-lead disadvantage (observation history loses predictive value as lead time grows, leaving TimesFM-3’s remaining information – the ensemble mean alone – no richer than what EMOS already uses). This explanation was conditional on whether ensemble spread could be added to TimesFM-3’s covariate channel without degrading it – a question this project has since tested directly (Section 5.4). The channel does accept a second, simultaneous input (Section 5.7), but supplying spread this way did not close the calibration deficit: CRPS and interval width both worsen, and coverage moves only marginally closer to the nominal target at a substantially wider interval. This does not confirm the information-asymmetry framing for this half of the mechanism question, but it does not confirm an architectural explanation either (Section 5.4, Section 6). Section 5 tests both halves of this explanation directly: the observation-history half by augmenting EMOS with a persistence predictor, and the calibration half by supplying TimesFM-3 with ensemble spread as a second covariate.

Two other covariate-aware zero-shot approaches make a different design choice than the frozen, built-in covariate channel this paper relies on: Auer et al. [2025] inject covariates through in-context examples rather than an architectural slot, requiring no model changes but also no guarantee the model attends to them the way a dedicated covariate channel would; Qin et al. [2025] instead adapt a pretrained TSFM’s covariate handling per-task through a small tuning step, which this paper’s zero-shot, no-fine-tuning design (Section 3.1) deliberately avoids.

¹Native past-future covariate support is documented for the 3.0 release specifically, not in the original TimesFM architecture paper [Das et al., 2024]: model card at <https://huggingface.co/google/timesfm-3.0-pytorch> and release notes at <https://research.google/blog/timesfm-3-a-zero-shot-foundation-model-for-multivariate-forecasting/> (Google Research; both accessed 2026-09-07).

Table 1: Information supplied to each method in this study – not each method’s inherent architectural capability. TimesFM-3’s “no” for ensemble spread reflects this study’s covariate design, not a confirmed limit of the 3.0 release (see note below and Section 5.7).

Method	Observation history	Ensemble mean	Ensemble spread
EMOS (pooled / local)	no	yes	yes
DRN	no	yes	yes
TimesFM-3 (zero-shot)	yes	yes	no*
Raw ensemble	no	yes (empirical)	yes (empirical)
Variance-inflation baseline	no	no	yes (scaled)

*Not supplied in this study. TimesFM-3’s covariate parameter accepts a second, simultaneous channel by design (see the note above), and a full evaluation supplying it found CRPS and interval width both worsen, not improve (Section 5.4).

3.3 Metrics

The Continuous Ranked Probability Score (CRPS) is our primary metric. Every reported CRPS value in this paper, for every method, is computed by the same pinball-loss-based quantile estimator [Jordan et al., 2019] applied to the identical 9-level quantile representation: EMOS and DRN’s own closed-form Gaussian CRPS [Gneiting and Raftery, 2007, Hersbach, 2000] is used only inside their training objective (Section 3.1), never for a reported evaluation metric, so no method receives an analytic-CRPS advantage over the discretized estimator every other method is necessarily scored with. We additionally report the tail-weighted CRPS (twCRPS), the same pinball-loss estimator restricted to the upper two quantile levels (0.8, 0.9) [Allen, 2024, Taillardat et al., 2023], as a check on tail behavior; Zamo and Naveau [2017], Bolin and Wallin [2023], and Arnold et al. [2024] discuss estimator choice and decomposition properties of CRPS more broadly, and Leutbecher and Haiden [2020] interprets CRPS differences directly. We additionally report *coverage@80%*: the empirical fraction of observations falling within the predicted p10–p90 interval, against a nominal level of 0.80 – not 0.90, since our fixed 9-level quantile grid has no p05/p95 pair to compute a 90%-nominal interval from; every mention of coverage in this paper refers to this 80%-nominal band and is labeled accordingly throughout. Finally we report *interval width* (the p10–p90 range in Kelvin) as a measure of sharpness. Per Gneiting et al. [2007]’s principle of maximizing sharpness subject to calibration, interval width is reported as a curve for every method and is never assigned a single “better” direction or a breakpoint of its own (Section 3.5): a narrower interval is only meaningful jointly with adequate coverage, not on its own. PIT histograms (Section 4) are likewise reported descriptively; Bröcker [2018] and Bröcker and Ben Bouallègue [2020] show that formal rank-uniformity hypothesis tests require correction for serial dependence, which our PIT values genuinely exhibit, so no such test is applied here (Section 5.7). Dirkson and Buehner [2026] further shows that coverage and rank-based reliability diagnostics can both be misled by the same covariance pathology; we do not claim these descriptive diagnostics are immune to that critique (Section 5.7).

3.4 Statistical significance

The evaluation set is a station $\times$ time $\times$ lead panel, not a single independent sample: instances share temporal autocorrelation within a station and spatial (synoptic) dependence across the 49 stations on a given day. A textbook Diebold–Mariano test [Diebold and Mariano, 1995, Diebold, 2015] assumes one autocorrelated series and is not directly applicable to this panel structure; its small-sample behavior is separately unreliable even in the single-series case it was designed for [Coroneo and Iacone, 2020]. We test two block definitions, aggregating the per-instance loss differential to one mean value per block before testing: a *station block* (absorbing

within-station temporal correlation, but not cross-station dependence) and a *day block* (absorbing cross-station synoptic dependence on a given day, following the logic of exact tests under contemporaneous correlation, Luger, 2004, but not multi-day persistence in the loss differential itself). We do not additionally test a multi-day moving-block variant that would absorb such short-range temporal persistence; the day block already targets the specific mechanism this analysis was designed to address (same-day synoptic systems affecting many stations at once), and we report this as a remaining, open block-definition question rather than as a claim that day-blocking is fully sufficient. For each block definition we report a paired *t*-test and a Wilcoxon signed-rank test on the block means, and a block-bootstrap confidence interval on the CRPS skill score (resampling whole blocks with replacement). We call this a *station-blocked* or *day-blocked paired test*, not a Diebold–Mariano test: the classical test’s assumptions do not transfer to this station $\times$ time $\times$ lead panel.

3.5 Breakpoint definition

The paper’s central quantity is the training-data size at which a trained method’s performance crosses a zero-shot reference’s. We define N days of training as $k = \text{round}(N/3.4833)$ issue-date cases, 3.4833 being the reforecast archive’s measured ratio of calendar days to sampled cases (Section 2). A breakpoint is the linearly interpolated case count k at which a trained method’s metric crosses the zero-shot reference’s value, computed over an ordered sequence of tested k ; we distinguish three outcomes explicitly rather than collapsing them into one “no breakpoint” label, which conflates opposite situations: *a crossing at $k = X$* (reported with its calendar-day equivalent), *already better at the smallest tested k* (no crossing needed because the trained method starts ahead), and *no crossing within the tested range* (the trained method remains behind throughout every tested k , which is a distinct and, for the zero-shot model’s defense, more informative outcome than a numerical breakpoint would be).

4 Results

4.1 Full-training-data comparison

At full training data (Table 2), every trained method beats TimesFM-3 on CRPS: EMOS pooled (1.2612 K), EMOS local (1.1023 K), and DRN (1.2251 K), against TimesFM-3’s 1.3592 K and the raw ensemble’s 1.3590 K. The trivial variance-inflation baseline beats TimesFM-3 on CRPS too: both the CRPS-optimal variant (a, 1.3005 K) and the fixed-climatological variant (b, 1.3042 K) undercut TimesFM-3’s 1.3592 K, and (b) needs no training data at all – a single zero-shot scalar rescaling of the raw ensemble’s spread already beats a frozen foundation model on this metric. On the calibration-sharpness axis the same trivial rescaling does not match TimesFM-3: the coverage-matched variant (c), calibrated on training data alone to reach as closely as a training-fit multiplier allows TimesFM-3’s own test coverage (0.8221 achieved against a 0.7603 target, Section 3), needs 6.7224 K to do so – substantially wider than TimesFM-3’s 5.1379 K, even after accounting for (c) landing closer to the nominal 0.80 target than TimesFM-3 itself, which alone would call for a somewhat wider interval. Variant (d), never a legitimate baseline (Section 3.1), bounds how much of that width gap could be a leakage artifact of (c)’s train/test split rather than a real limit: fit directly on test-set coverage, it reaches 1.3054 K on CRPS and 0.7603 achieved against the identical 0.7603 target – matched exactly, not merely approximately – yet still needs 5.5344 K, wider than TimesFM-3’s 5.1379 K even at this leakage-optimistic upper bound. This margin is not just a point estimate: a day-blocked bootstrap on the per-instance width difference gives 0.3965 K (95% CI [0.2887, 0.4997]), entirely positive – the leakage-optimistic variant is significantly wider than TimesFM-3, not merely wider on average. The bootstrap resamples day-blocks around a single leaky multiplier fit once on the full matched test set; the interval is conditional on that multiplier and quantifies sampling variability in the width difference, not

Table 2: Full-training-data (N=full, 4180 cases) comparison. CRPS, twCRPS, and interval width in Kelvin; coverage against the 80%-nominal p10–p90 band. DRN is trained under a feature-constrained setup (Section 5.7), not its originally-intended full feature set.

Method	CRPS (K)	twCRPS (K)	Coverage @80%	Interval width (K)
Raw ensemble	1.3590	1.1109	0.4259	2.3716
Var.-inflated, CRPS-optimal (a)	1.3005	0.9522	0.7388	5.2037
Var.-inflated, fixed $\lambda = 1.5$ (b)	1.3042	0.9776	0.6096	3.7258
Var.-inflated, coverage-matched (c)	1.3361	1.0029	0.8221	6.7224
Var.-inflated, leakage-optimistic (d) [†]	1.3054	0.9579	0.7603	5.5344
TimesFM-3 (zero-shot)	1.3592	1.0268	0.7603	5.1379
TimesFM-3, two-covariate (mean + spread) [‡]	1.3837	–	0.8027	5.8246
EMOS pooled	1.2612	0.9138	0.8303	5.4559
EMOS local	1.1023	0.8239	0.8289	4.7548
DRN	1.2251	0.8989	0.7720	4.6494

[†]Never a legitimate baseline: λ is fit directly on test-set coverage (Section 3.1), an upper bound on what any trivial rescaling could achieve at this coverage target, not a leakage-free result.

[‡]Not this paper’s main TimesFM-3 configuration (that is the zero-shot row above, used everywhere else in this paper): an additional experiment supplying ensemble spread as a second covariate (Section 5.4).

twCRPS is omitted (–) because per-instance quantiles for this run were not persisted; only the aggregate metrics shown were saved.

variability in the multiplier itself. TimesFM-3’s real advantage over one-parameter rescaling is therefore narrower than a CRPS-only reading suggests, and it sits on a different axis than CRPS: on CRPS, a trivial multiplier already beats it, zero-shot; the width/coverage trade-off is where TimesFM-3 holds an edge that neither a leakage-free nor a leakage-optimistic scalar rescaling reproduces. Section 5 returns to what this split result means for claims that trivial baselines match foundation models generally [Zhang and Gilpin, 2025].

A separate experiment supplies TimesFM-3 with ensemble spread as a second, simultaneous covariate (Table 2’s two-covariate row). CRPS worsens to 1.3837 K and coverage moves closer to the 0.80-nominal target (0.8027 against 0.7603) at a substantially wider interval (5.8246 K) – a shift along the calibration-sharpness trade-off, not a clean improvement, discussed fully in Section 5.4.

4.2 The training-data-size breakpoint curve

Figure 1 shows the paper’s headline result: EMOS crosses TimesFM-3’s CRPS at 1.6 cases pooled (6 calendar days) and 1.8 cases locally (6 days). That raw crossing is a point estimate, not itself statistically significant: testing significance at every low- N point on this curve ($k = 1, 2, 3, 5, 7, 9$, Section 4.5) shows the advantage is significant from 3 cases (10 calendar days) onward for pooled EMOS and from 3 cases (10 days) for local EMOS – under day-blocking, our primary block definition; the station-blocked t -test does not reach significance anywhere in the tested $k = 1$ – 9 range for either variant, while the station-blocked Wilcoxon test does, from $k = 2$ onward – the same test-statistic divergence already reported at 9 cases (Section 4.5), now shown to persist across the whole low- k range. Both numbers matter and neither replaces the other: the raw crossing (6 days) is where the two methods’ CRPS point estimates change order; the day-blocked significant crossing (10 days) is where that difference is established rather than plausibly noise. DRN needs 15.1 cases (52 days) – under the feature-constrained setup this paper trains it in, not DRN’s originally-intended full feature set (Section 5.7). The variance-inflation baseline (variant a, CRPS-optimal, refit per training size the same way EMOS is) is already

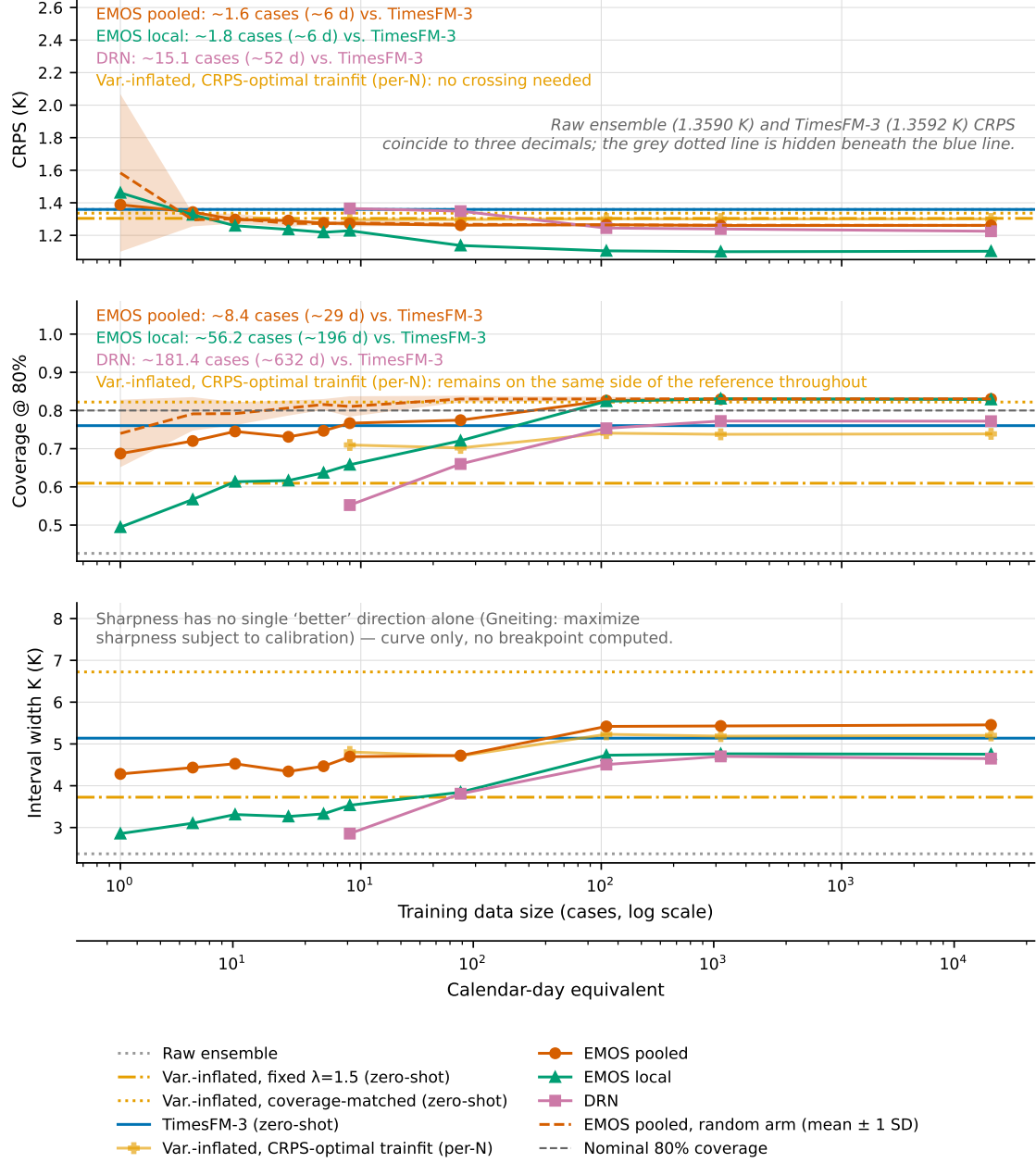


Figure 1: CRPS, coverage@80%, and interval width against training data size (cases, log scale, with calendar-day equivalent on the secondary axis). Horizontal lines are N-independent zero-shot references; curves are trained methods refit at each N.

better at the smallest tested N : at 30 days it already reaches 1.2972 K, below TimesFM-3’s 1.3592 K, and stays there throughout the tested range – the same qualitative pattern Table 2 shows at full data (Section 4.1). Coverage tells a different story (ratios below computed from the unrounded breakpoint estimates, not the values as displayed): EMOS pooled crosses TimesFM-3’s coverage at 8.4 cases (29 days) – 5.1x later than its CRPS crossing – and EMOS local not until 56.2 cases (196 days), 32.0x later than its own CRPS crossing; DRN crosses at 181.4 cases (632 days). Interval width has no crossing at all (Section 3.5): EMOS pooled’s interval starts narrower than TimesFM-3’s and crosses to wider somewhere between 90 and 365 training days, while EMOS local stays narrower than TimesFM-3 across the entire tested range even as it reaches comparable coverage – a sharpness advantage a pooled fit does not share.

4.3 Lead-time resolution

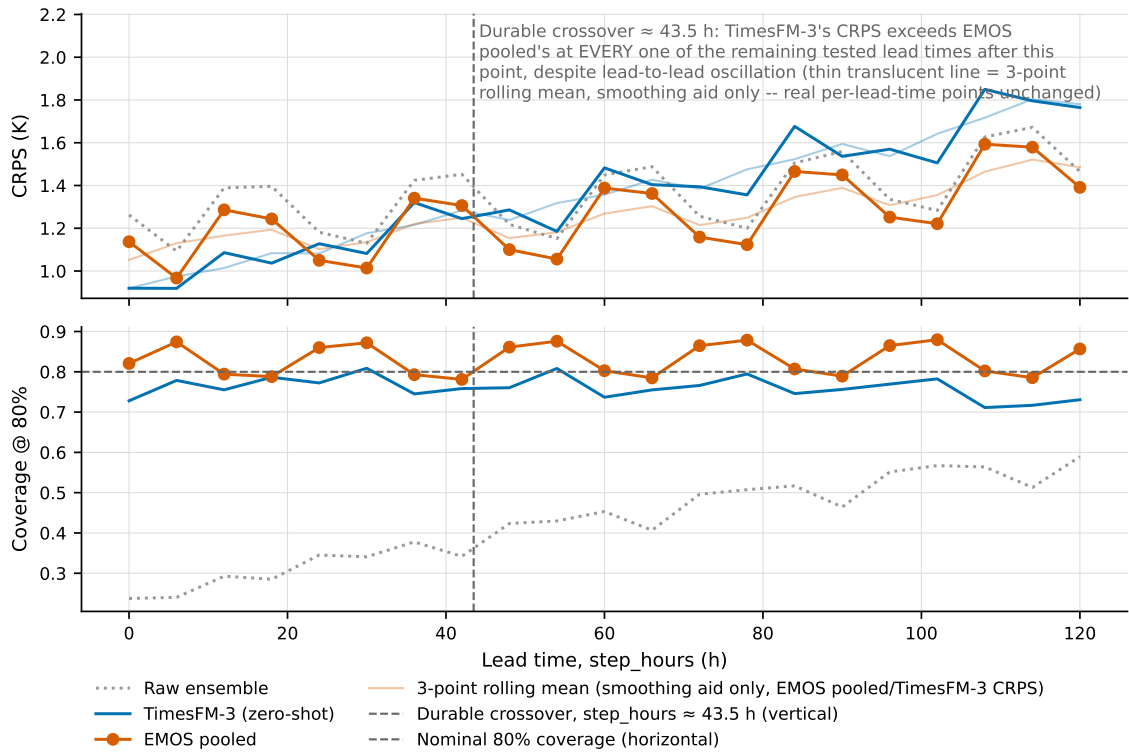


Figure 2: CRPS and coverage@80% by lead time (21 steps, 0–120 h).

Pooling over all 21 lead times hides a reversal (Figure 2): TimesFM-3’s CRPS advantage is real but short-lived, with a first sign flip at 22.36 h and a durable crossover at 43.49 h, after which TimesFM-3 remains the worse method at every longer tested lead time. Recomputing the breakpoint curve separately within three lead-time buckets (0–24 h, 24–72 h, 72–120 h; Figure 3) sharpens this into the paper’s most lead-dependent result: at 72–120 h, both EMOS pooled and EMOS local already beat TimesFM-3 on CRPS at the smallest tested training size (a single case); at 0–24 h, pooled EMOS does not cross TimesFM-3’s CRPS anywhere in the tested range, from 1 to 4180 cases. Coverage crossings follow the same ordering (earliest at long lead, latest at short lead) but at substantially larger training sizes than CRPS in every bucket.

4.4 Sharpness, calibration, and PIT diagnostics

Figure 4 makes Section 4.1’s width/coverage trade-off visible directly: EMOS local’s trajectory sits up and to the left of every other trained method’s, reaching comparable coverage to

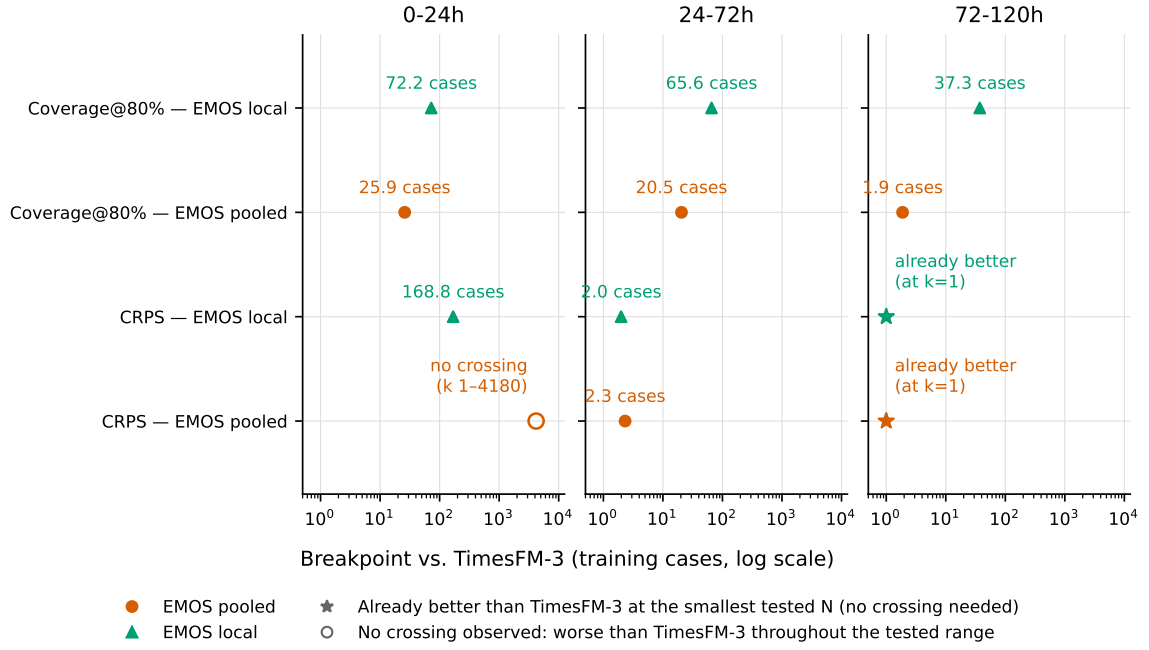


Figure 3: Training-data-size breakpoint, computed separately within each lead-time bucket, for CRPS and coverage@80%.

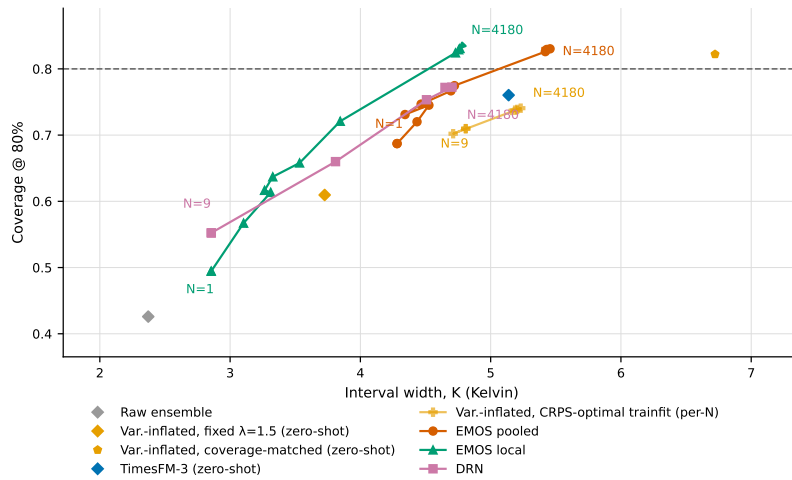


Figure 4: Interval width against coverage@80% for every method; trained methods shown as trajectories across training size, zero-shot methods as single points.

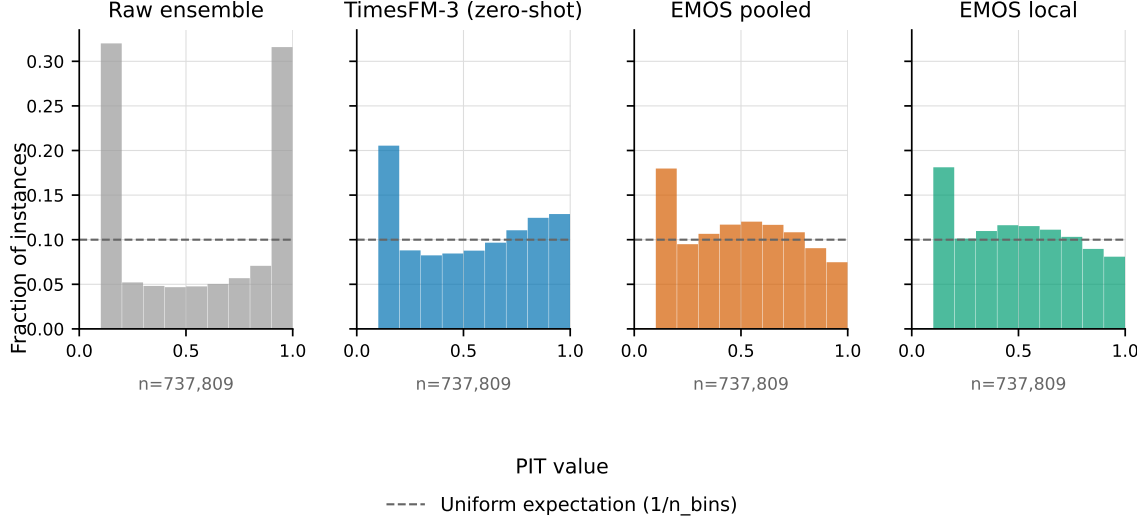


Figure 5: PIT histograms at full training data. PIT uniformity is assessed descriptively; no formal uniformity test valid under serial dependence is applied.

TimesFM-3 with a narrower interval throughout. The raw ensemble’s PIT histogram (Figure 5) shows the classic severe-underdispersion signature – roughly a third of observations fall in each of the two outermost bins – consistent with its 0.4259 empirical coverage against a 0.80 nominal target; TimesFM-3 and both EMOS variants show comparatively flat histograms, read descriptively per Section 3.

Table 2’s twCRPS column follows the same ranking as CRPS itself for every method (e.g. EMOS local’s 0.8239 K remains the lowest, TimesFM-3’s 1.0268 K remains among the highest) – this checks the upper end of the quantile grid this study fixed for every method, not whether TimesFM-3 genuinely lacks tail behavior beyond p90 (Section 3.3).

4.5 Statistical significance

At the pooled headline comparison (EMOS pooled vs. TimesFM-3 at 9 cases, 30 days), the CRPS advantage is robust to block definition: station-blocked and day-blocked tests agree in direction, and the day-blocked test – which additionally absorbs cross-station synoptic dependence the station block cannot – gives $p = 1.50e - 12$ (paired t) and $p = 3.59e - 06$ (Wilcoxon), both far below the station-blocked test’s already-significant Wilcoxon result ($p = 1.88e - 05$) and its non-significant t -test ($p = 0.1485$). The coverage advantage does not clear the same bar: station-blocked (t : $p = 0.7432$; Wilcoxon: $p = 0.0165$) and day-blocked (t : $p = 0.0506$; Wilcoxon: $p = 0.4391$) tests disagree with each other on both significance and, at the day block, direction of evidence between the two test statistics. We do not report EMOS as coverage-superior to TimesFM-3 at this training size: the honest summary is that the CRPS result is significant under every tested block definition and test statistic, while the coverage result is not.

The breakpoint curve’s own raw crossing (Section 4.2: 1.6 cases pooled, 1.8 cases locally) is a different, smaller training size than 9 cases, and was not itself significance-tested until now. Repeating the day- and station-blocked tests at every point on the low- N breakpoint grid ($k = 1, 2, 3, 5, 7, 9$) shows the CRPS advantage is not significant at the raw crossing: the earliest k at which both the day-blocked t -test and Wilcoxon test (day-blocking is our primary block definition) agree the advantage is significant is 3 cases (10 days) for pooled EMOS (day-blocked t : $p = 5.99e - 07$; Wilcoxon: $p = 0.0186$) and 3 cases (10 days) for local EMOS (day-blocked t : $p = 7.51e - 13$; Wilcoxon: $p = 1.03e - 09$) – meaningfully later than the raw crossing, though still

a small absolute training size. The station-blocked paired t -test is not significant at this same k for either variant (pooled: $p = 0.3446$; local: $p = 0.0529$), while the station-blocked Wilcoxon test is (pooled: $p = 6.81e - 05$; local: $p = 3.49e - 08$) – the same station-block t -vs-Wilcoxon divergence already seen at 9 cases above holds throughout the tested range, not just at that one point; the station-blocked t -test is not significant at any k we tested from 1 to 9 cases for either EMOS variant.

Repeating both block definitions separately within each lead-time bucket, using full-training-data EMOS pooled against TimesFM-3, resolves this divergence differently per bucket rather than uniformly. At 0–24 h, the CRPS gap is significant in TimesFM-3’s favor (day-blocked t : $p = 4.29e - 61$; station: $p = 0.0354$). The coverage gap in the same bucket is significant in EMOS’s favor instead (day-blocked t : $p = 8.40e - 53$; station: $p = 0.0255$) – EMOS is significantly *closer to nominal coverage* even in the one bucket where TimesFM-3 wins on CRPS. At 24–72 h, station- and day-blocked CRPS tests disagree on significance (day: $p = 4.33e - 07$; station: $p = 0.2268$), with the day-blocked test – our primary block definition – favoring EMOS. At 72–120 h, both block definitions agree EMOS’s CRPS advantage is significant ($p = 3.69e - 35$ day-blocked, $p = 3.31e - 08$ station-blocked). Across all three buckets, EMOS’s coverage advantage over TimesFM-3 is significant under day-blocking (p between $8.40e-53$ and $3.06e-64$). The pattern across every test in this section is consistent in direction and variable in magnitude and significance, not uniform: we report each result at the resolution it was actually tested, rather than a single pooled significance claim standing in for all of them.

5 Discussion

5.1 Why a few days is enough

The paper’s central practical result is that EMOS pooled crosses TimesFM-3’s CRPS at 1.6 cases (6 calendar days, Section 4.2) – but that raw crossing is a point estimate, not yet a statistically defensible claim. Testing significance across the low- N range shows the earliest training size at which the advantage is significant under both a day-blocked t -test and a Wilcoxon test is 3 cases (10 days, Section 4.5) – day-blocking being our primary block definition; the station-blocked t -test never reaches significance in the tested $k = 1$ –9 range, while the station-blocked Wilcoxon test does from $k = 2$ onward, the same divergence already reported at 9 cases now shown to persist throughout – still a small absolute training size, and still well inside the regime Baran and Lerch [2018] and Gebetsberger et al. [2018] already identify as usable for station-adaptive EMOS, and that the time-series and lead-time-continuous EMOS variants of Jobst et al. [2024] and Wessel et al. [2024] target directly – but nearly twice the raw crossing’s calendar-day figure, not the same number. The practical implication for a newly commissioned station is direct, stated at the number the evidence actually supports: roughly 10 days of paired data, not months, are enough to stop relying on a zero-shot foundation model’s raw CRPS with statistical confidence; the raw crossing at 6 days is the earliest the two methods’ CRPS point estimates change order, not the earliest that difference is established rather than noise.

This result is deliberately scoped to CRPS alone. Coverage tells a different story at substantially more data – 5.1x later for pooled EMOS, 32.0x later for local EMOS (ratios from the unrounded breakpoint estimates, not the values as displayed; Section 4.2: 8.4 cases pooled, 56.2 cases locally), and interval width has no single crossing point at all – EMOS pooled starts sharper than TimesFM-3 and crosses to wider as it is given more data, while EMOS local stays sharper throughout the tested range (Section 3.5’s three-outcome distinction exists precisely so this kind of metric-specific divergence is reported as what it is, not flattened into one number). "A few days is enough" is therefore true for CRPS specifically, not for every metric this paper reports – Sections 5.2 and 5.3 address what CRPS alone does and does not establish about why.

The question this paper asks – how much training data a trained method needs to replace a zero-shot foundation model – has a deployment-cost dimension this paper has not systematically

measured: TimesFM-3 requires a GPU forward pass through a 330-million-parameter pretrained model for every inference call, while fitting EMOS end-to-end on the full training archive took 109.8 s (pooled) or 146.7 s (local) of CPU time in this study’s own runs, and inference afterward is a closed-form Gaussian evaluation, not a further optimization. Once the training-data threshold in this paper is crossed, the practical choice is not a single-axis trade-off between accuracy and convenience: the trained method is both more accurate and cheaper to run.

5.2 Positioning against trivial baselines

Section 4.1’s full-data comparison (Table 2) shows a result that splits by metric in a way worth stating plainly rather than softening: on CRPS, a one-parameter variance-inflation rescaling of the raw ensemble already beats TimesFM-3, and the fixed-climatological variant (b, 1.3042 K vs. TimesFM-3’s 1.3592 K) does so with no training data at all – squarely the pattern Zhang and Gilpin [2025] describes for trivial baselines against zero-shot foundation models generally, now confirmed in station postprocessing specifically. A comparable dynamic – a pretrained model matching, but not decisively beating, a much simpler well-tuned alternative – is reported in a different domain by Molinaro et al. [2026].

On the calibration-sharpness axis, the evidence points the other way, and this finding is now settled rather than conditional: the coverage-matched variant (c), its multiplier fit on training data alone (never on test-set coverage, Section 3), needs 6.7224 K to reach as close to TimesFM-3’s own coverage as a training-fit multiplier allows – substantially wider than TimesFM-3’s 5.1379 K. Variant (c) is leakage-free by design, which means it is not an upper bound on what a trivial one-parameter rescaling could achieve at TimesFM-3’s coverage target – only a lower bound on how well a *legitimately deployable* one can do. The leakage-optimistic counterpart – a multiplier fit directly on the test set’s own coverage, answering "what is the best any trivial rescaling could ever do here, even with an unfair advantage" – needs 5.5344 K, still wider than TimesFM-3’s 5.1379 K, at matched coverage: the leaky variant’s root-finder converges to 0.7603, identical to TimesFM-3’s 0.7603 target to within floating-point precision, so the width comparison is not confounded by a residual coverage gap the way variant (c)’s is. This margin is not a bare point estimate: a day-blocked bootstrap on the per-instance width difference gives 0.3965 K (95% CI [0.2887, 0.4997]) – entirely positive, so the difference is statistically distinguishable from zero, not merely numerically positive on average. The interval is conditional on the leakage-optimistic multiplier fit once on the full test set, not re-fit within each bootstrap resample: it quantifies sampling variability in the width difference at that fixed multiplier, not variability in the multiplier itself. This margin is also not explained by EMOS’s own training/test ensemble-size mismatch: subsampling the test side down to the training side’s member count changes EMOS’s own interval width by no more than 0.0171 K (Section 5.7), under one-tenth of this margin. TimesFM-3’s calibration-sharpness advantage is therefore real: it survives even this unrealistically favorable trivial baseline, with uncertainty quantified (subject to that conditioning), so Zhang and Gilpin [2025]’s critique, while correct on CRPS (above), does not extend to this axis.

The honest summary of Section 4.1 is: TimesFM-3 loses to a trivial rescaling on CRPS, unconditionally; it beats even a leakage-optimistic trivial rescaling on sharpness at matched coverage, so the calibration-sharpness advantage is not a leakage-free-vs.-leaky comparison artifact.

5.3 The short-lead advantage: information asymmetry or architecture

This section remains provisional; the architectural-advantage hypothesis has not been confirmed. Section 3.2 proposed a single mechanism for TimesFM-3’s short-lead CRPS advantage (Section 4.3: first sign flip at 22.36 h, durable crossover at 43.49 h): access to two channels EMOS and DRN were not given in this study’s configuration (Table 1) – the station’s

own recent observation history, and ensemble spread. These are two separate questions with two separate tests; one has now been run, with a limited, inconclusive result, and one is still outstanding.

Observation history (tested; does not support the information-asymmetry explanation, but does not settle the question either). A persistence-augmented EMOS at 0–24 h tests whether adding the station’s own last observation as a predictor closes the CRPS gap toward TimesFM-3. It does not, substantially: only 3.4% of the gap closes (persistence weight 0.0786, matched subset, 140,457 instances), confirmed after ruling out lead-0 contamination as a confound (Section 5.7: exact-match and trend checks both clean). This result does not support the information-asymmetry explanation as stated – the simplest way of giving EMOS the missing observation-history channel does not close the gap. It does not, on its own, establish that the advantage is architectural instead: the test has three real limitations that keep it from distinguishing the two hypotheses cleanly, not just describing them as a formality.

- *Lead-pooled fit.* EMOS pools all 21 lead times into one fit (Section 3.1); the persistence term is fit once, lead-blind, inside that pooled model, while TimesFM-3’s actual covariate and target information vary naturally with lead time. The two methods are not given the same structural opportunity to use lead-dependent information – this is not an equal comparison.
- *One linear term, one lag.* The persistence predictor is a single linear term on the single most recent observation. TimesFM-3 receives a context window of 40 or more past observations, processed by a sequence model that can learn arbitrarily nonlinear, multi-lag structure. Adding one linear lag does not give EMOS "comparable information" to a 40-observation context in any meaningful sense – only a much smaller, cruder approximation to it.
- *Metric-dependent.* In the same 0–24 h bucket, TimesFM-3 wins on CRPS but EMOS is significantly *closer to nominal coverage* (day-blocked t : $p = 8.40e - 53$, EMOS’s favor, Section 4.5). Whatever "architectural advantage" this test might support holds for CRPS specifically, not uniformly across every metric in the same lead-time bucket.

Given these three limitations, the honest reading of this test is narrower than "architecture, not information asymmetry": it shows that the simplest information-asymmetry fix is insufficient, not that architecture is confirmed as the explanation. A test able to distinguish the two hypotheses would need a lead-resolved EMOS fit (not pooled) with a multi-lag persistence predictor (not one linear term on one lag) – this paper does not run that test, and does not claim the weaker test run here settles the question.

Ensemble spread / calibration (resolved: does not support the asymmetry framing). The other half of the proposed asymmetry – that TimesFM-3’s calibration deficit relative to EMOS reflects a missing spread channel, not an architectural limit – has now been tested directly. The model’s covariate channel does accept spread as a second, simultaneous input, and the model is genuinely sensitive to that channel’s real informational content, not merely its presence (a preliminary test confirms both, Section 3.2, Section 5.7), but supplying it this way did not close the calibration deficit: CRPS and interval width both worsen, and coverage moves only marginally closer to nominal at a substantially wider interval (Section 5.4 reports the full result). This does not confirm the asymmetry framing for this half of the mechanism question, and the observation-history result above does not settle it either: the two channels test genuinely different things, and a finding for one is not evidence for the other.

5.4 The two-covariate evaluation: does spread change calibration?

Section 3.2’s preliminary test (T1) established that TimesFM-3’s covariate channel accepts ensemble spread as a second, simultaneous input alongside the ensemble mean, and that the model’s

output is genuinely sensitive to that channel’s real informational content. This section reports the full-scale evaluation that preliminary result motivated: supplying both the ensemble mean and ensemble spread as simultaneous past-future covariates, on the same 737,809 matched instances used for every other TimesFM-3 number in this paper, and comparing the result against the mean-only configuration used everywhere else.

Supplying spread did not improve TimesFM-3’s CRPS: it worsens from 1.3592 K to 1.3837 K, a net negative of 0.0246 K. No uncertainty interval has been computed for this difference; its magnitude is comparable to the ensemble-size sensitivity measured directly in Section 5.7 (0.0121–0.0141 K), so it should be read as a real but not dramatically large effect, not a precisely bounded one. Coverage moves closer to the 0.80-nominal target, from 0.7603 to 0.8027: in practice, the two-covariate configuration is calibrated at this target, where the mean-only configuration under-covers. Interval width also increases, from 5.1379 K to 5.8246 K (0.6867 K wider) – but by this paper’s own standard of judging sharpness jointly with calibration (Section 3, Gneiting et al., 2007), these two widths are not directly comparable on the same axis: they are measured at different coverage levels, and a wider interval is the expected cost of a configuration that is closer to properly calibrated, not on its own evidence of a worse model. This reading has independent support: fitting the leakage-optimistic multiplier (Section 5.2) at this same 0.8027 target, rather than TimesFM-3’s original 0.7603, still needs 6.3026 K – 0.4780 K wider than the two-covariate configuration’s own 5.8246 K at the identical coverage. Finding 2’s width advantage therefore holds at two independent coverage points, not only at TimesFM-3’s original one.

Supplying ensemble spread as a second covariate therefore worsens TimesFM-3’s CRPS and moves it along the calibration-sharpness trade-off – closer to nominal coverage, at a wider interval – rather than cleanly closing the calibration deficit: this is a shift along that trade-off, not a clean improvement or a clean failure. This result does not confirm the information-asymmetry framing for the calibration half of the mechanism question (Section 3.2): TimesFM-3 was not simply prevented by an input-design choice from recovering EMOS-like calibration once that choice was corrected. But it does not confirm architecture as the explanation instead, either: the model accepted the second input and was demonstrably sensitive to it (Section 5.7), and did shift its output along the calibration-sharpness trade-off when given it – it just did not convert that shift into a net improvement on CRPS, or into a clean win on the paper’s joint sharpness-calibration standard, within this study’s zero-shot, frozen-weights setup. Whether a fine-tuned or differently prompted configuration would do better remains untested and outside this paper’s scope; so does whether a different input scale or normalization convention for the spread channel would change this result (Section 5.7).

5.5 Implementation pitfall: covariate wiring

Kreusel et al. [2026] reports that covariate handling, not architecture or scale, is the dominant factor in time series foundation model performance on real tasks. This project’s own development history is a direct, first-hand instance of that finding rather than a citation of it: an earlier version of TimesFM-3’s covariate pipeline (Section 3.2) supplied an incoherent covariate channel – one that ran without error and produced plausible-looking quantile output – and measurably degraded TimesFM-3’s apparent skill before the bug was found and corrected. The pipeline did not crash; it silently produced a worse model. The practical warning for anyone implementing a covariate-aware time series foundation model pipeline is that "it runs and returns sensible-looking numbers" is not evidence the covariate is wired correctly – only a dedicated correctness check on the covariate channel itself (not just on the model’s output shape or its overall CRPS) would have caught this before it silently degraded results. This experience is the direct reason this project later required a distinguishing correctness check – not just "does the call succeed," but "is the output genuinely sensitive to this covariate’s real informational content, not merely its presence" – before trusting any further covariate-channel change (Section 5.4’s ablation test

is that check, run precisely because of the lesson learned here).

5.6 The value of seasonal coverage in small training sets

Section 2’s two sampling arms answer different questions by construction: the contiguous arm’s small- k subsets see less seasonal variety than a full archive would, while the random arm’s uniform draw gives artificial full-season coverage even at small k . At 30 days, EMOS pooled’s coverage is already visibly different between the two arms – 0.8105 under the random arm against 0.7668 under the contiguous arm – while CRPS is nearly identical (1.2760 K vs. 1.2720 K). This is consistent with Section 5.1’s CRPS-crosses-fast, coverage-crosses-slow pattern having a seasonal-coverage component: a model can already predict the mean well from a handful of contiguous days, but calibrating the predictive interval correctly needs to have seen a representative spread of conditions, not just recent ones. Operationally, this means the realistic deployment scenario – a newly commissioned station with only contiguous recent history, the contiguous arm’s premise – reaches good CRPS quickly but should not be assumed to be well calibrated at the same training size a full-season random sample would be. Baran and Mihalina [2026] targets the same sharpness axis on the same benchmark; the overlap is in the axis studied, not in this specific contiguous-vs-random comparison, which does not appear in that work.

5.7 Limitations

Covariate channel design; tested, but does not resolve the mechanism question.

TimesFM-3’s covariate channel was given only the ensemble mean in this study’s main comparison (Section 3.2); whether supplying ensemble spread too would change calibration was tested directly (Section 5.4). The channel does accept a second, simultaneous input, and the model is genuinely sensitive to its real informational content, not merely its presence – but CRPS worsens when spread is supplied this way, and while coverage moves closer to nominal (a genuine calibration shift, not merely a wider interval for its own sake), the interval widens substantially to get there, so the net result is a shift along the calibration-sharpness trade-off, not a clean win (Section 5.4). This result does not confirm the information-asymmetry framing (spread was not simply withheld from a channel that would otherwise have used it straightforwardly beneficially), but it does not confirm architecture as the explanation either: the channel accepted the input and the model was demonstrably sensitive to it, and did shift its behavior in response to it, without that shift converting into a net improvement. Whether a fine-tuned or differently prompted configuration would use spread more successfully is untested and outside this paper’s scope. We report this as an unresolved mechanism question rather than resolving it either way beyond what this evidence supports.

Spread channel scale and normalization. Ensemble spread was supplied to TimesFM-3’s second covariate channel as a standard deviation, in the same physical units (Kelvin) as the ensemble mean but roughly two orders of magnitude smaller in raw value (typically 1–3 K vs. 270–290 K) – a real scale disparity between the two channels as supplied. This is not, on its own, an unaddressed confound: TimesFM-3’s own preprocessing z-normalizes each covariate channel independently, computing a running mean and standard deviation per channel rather than sharing statistics across channels (confirmed directly from the installed `timesfm3` package’s source, not inferred), so the raw magnitude difference between the mean and spread channels does not by itself explain why supplying spread failed to improve performance. The spread channel was supplied at a single scale and as a single channel, per-channel normalization is confirmed rather than merely assumed, and the result above should be read as "this specific input design does not help," not as "TimesFM-3 cannot use ensemble spread" more broadly – untested alternatives include a differently transformed spread channel (e.g. log-spread) or a channel expressing spread relative to the mean.

Lead-time pooling. EMOS and DRN each fit a single parameter set pooled across all

21 lead times (Section 3.1), while TimesFM-3 and the raw ensemble vary naturally by lead time. A lead-blind fit is necessarily too wide at short lead (where true uncertainty is small) and too narrow at long lead (where it is large) relative to a lead-time-aware fit, which means the CRPS crossings reported in Section 4.2 are, if anything, understated for EMOS and DRN, not artifacts of this pooling. It is a genuine limitation for the sharpness comparison specifically, since a lead-blind interval’s width behavior is not directly comparable to TimesFM-3’s per-lead-varying output. For DRN in particular, this feature-constrained setup is not the architecture’s originally-intended one (Rasp and Lerch, 2018 additionally use auxiliary NWP fields and lead time as inputs): where DRN underperforms in this paper, the correct reading is that this architecture, under this feature-constrained setup, was outperformed – not that the architecture itself is weak.

Persistence test run on a lead-pooled EMOS fit. The persistence-augmented EMOS test (Section 5.3) inherits this same lead-pooling limitation directly: the persistence predictor is fit once, lead-blind, inside a model already disadvantaged by pooling across all 21 lead times against a naturally lead-varying TimesFM-3. This reduces the test’s power to distinguish an information-asymmetry explanation from an architectural one – a lead-resolved EMOS fit with a multi-lag persistence predictor would be a meaningfully stronger test, and this paper does not run one.

Target-variable specificity. The breakpoint measured here is for 2 m temperature, where EMOS’s Gaussian predictive family is close to ideally specified. For targets requiring non-Gaussian or target-specific reformulations (Section 1: e.g. Scheuerer, 2013’s precipitation-specific EMOS), the trained method’s small-sample advantage would not necessarily appear as early.

Tail truncation in twCRPS. The tail-weighted CRPS defined in Section 3.3 uses the upper two levels of a fixed 9-level quantile grid (0.8, 0.9) as its tail weighting. This tests the upper end of the quantile grid this study fixed for every method, not whether TimesFM-3 genuinely lacks tail behavior beyond p90 – the grid itself has no quantile levels past 0.9 for any method, so no comparison in this paper can distinguish "no tail" from "a tail this grid cannot see."

PIT diagnostics are descriptive only. Bröcker [2018] and Bröcker and Ben Bouallègue [2020] show that formal rank-uniformity hypothesis tests require correction for serial dependence, which this study’s PIT values genuinely exhibit (station $\times$ time $\times$ lead panel structure, Section 3.4); no such test is applied to the PIT histograms in Section 4.4, which are read descriptively for this reason. Dirkson and Buehner [2026] further shows that coverage and rank-based reliability diagnostics can both be misled by the same covariance pathology – we do not claim the descriptive PIT reporting in this paper is immune to that critique; it is a genuine, unresolved limitation, not one worked around by reporting PIT histograms instead of a formal test.

Random-arm seed count. The random sampling arm’s mean and standard deviation (Section 2) are estimated from 5 seeds. This is enough to see the arm’s qualitative pattern relative to the contiguous arm, but the standard deviation itself is estimated noisily from so few draws – it should be read as an indicator of spread, not a precise quantity, and Figure 1’s random-arm error bars inherit this same noise.

Ensemble size mismatch. EMOS’s training-side ensemble variance is estimated from 11 members against 51 on the test side (Section 2); an 11-member variance estimate is intrinsically noisier than a 51-member one of the same true spread, so EMOS is trained against a noisier variance estimate than the one used to evaluate it. This is no longer an unbounded concern: subsampling 11 of the test side’s 51 members (10 draws, recomputing the ensemble mean/variance from just those 11 and re-evaluating the already-fitted full- N EMOS models) changes pooled EMOS’s own CRPS by 0.0121 K, its interval width by 0.0131 K, and its coverage@80% by 0.0051 (local EMOS: 0.0141 K, 0.0171 K, 0.0078 respectively) – both width differences are under one-tenth of Section 5.2’s 0.3965 K leakage-optimistic-width margin. The ensemble-size mismatch is real and measurable, but it is too small to be a material contributor to that finding.

Unverified chronological assumption. The contiguous sampling arm orders cases by

descending (`year_idx`, `time_idx`), assuming ascending `year_idx` is chronologically more recent (Section 2). Only `time_idx`'s ordering was independently verified against real decoded dates; `year_idx`'s polarity was not. This bounds its impact to training sizes beyond the 209-date single-cycle boundary ($k > 209$, i.e. N greater than roughly 728 calendar days) – below that boundary, every contiguous-arm result in this paper is unaffected regardless of which polarity is correct.

Training-data contamination risk. This study does not verify that TimesFM-3's pre-training corpus excludes the 2017–2018 test period used here. Data contamination in time series foundation model evaluation is an active, documented concern (Li et al., 2026, Chiazagomekpere, 2026, Balloccu et al., 2024), and benchmark-methodology work (Aksu et al., 2024, Meyer et al., 2025) argues evaluation protocols in this area have not caught up with the risk. We cannot rule this out for TimesFM-3 on EUPPBench; a demonstrated case of the same risk materializing in a different domain is given by Pan and Ezzat [2026]. If TimesFM-3's apparent zero-shot skill benefits from contamination, its true zero-shot CRPS would be worse than reported here, which would only strengthen this paper's conclusion that trained methods overtake it quickly – it does not threaten the paper's central claim, but it is a real, unaddressed limitation of the TimesFM-3 numbers taken in isolation.

Lead-0 contamination, checked and ruled out. Section 5.3 depends on the 0–24 h bucket, which includes `step_hours = 0` – the analysis time, where the "forecast" could in principle be tightly coupled to the observation it is verified against for every method, not only TimesFM-3. Two checks rule this out: the exact-match fraction between the ensemble-mean forecast and the observation at lead 0 is 0.000000 (no higher than the 0.000028 seen at any other lead, both consistent with floating-point noise, not literal copying); and lead 0's residual standard deviation (1.7644 K) is in line with what a linear trend fit to leads ≥ 6 h predicts at `step_hours = 0` (1.7387 K, ratio 1.0148) – if anything very slightly above the trend, not a discontinuous tightening below it. Every lead-time-resolved result in this paper that includes lead 0 (Section 4.3) can be read without this particular caveat.

6 Conclusion

At full training data, every trained method – EMOS pooled, EMOS local, and DRN – beats TimesFM-3's CRPS, and a one-parameter variance-inflation rescaling of the raw ensemble does too, with a fixed-climatological variant achieving this using no training data at all (Section 4.1). This advantage appears at very small training sizes: EMOS pooled's raw CRPS crossing is at 1.6 cases (6 calendar days), and that crossing becomes statistically significant, under day-blocking (our primary block definition), from 3 cases (10 days) onward for both EMOS variants – the station-blocked t -test does not reach significance anywhere in the tested $k = 1$ –9 range, while the station-blocked Wilcoxon test does from $k = 2$ onward (Section 4.5). For a newly commissioned station, this is the practical answer to this paper's title question on CRPS: roughly 10 days of paired ensemble/observation history, not months, are enough to reach a statistically defensible CRPS advantage over a frozen, zero-shot foundation model, and does so at a lower inference cost too, on the qualitative comparison in Section 5.1 (a real, measured EMOS fit time against an unmeasured but qualitatively larger GPU inference cost for TimesFM-3, not a quantified comparison on both sides). Coverage crosses far more slowly than CRPS – 5.1x later for pooled EMOS, 32.0x later for local EMOS – so this practical answer holds for CRPS specifically, not for calibration.

On the calibration-sharpness axis, TimesFM-3 holds a real advantage over a trivial rescaling, and this is settled, not conditional: even a leakage-optimistic multiplier fit directly on test-set coverage – an upper bound no legitimately deployable trivial rescaling could exceed – needs 5.5344 K, still wider than TimesFM-3's 5.1379 K at matched coverage, by a margin of 0.3965 K (95% CI [0.2887, 0.4997], day-blocked bootstrap, Section 5.2) that excludes zero entirely. This advantage is not an artifact of comparing a leakage-free multiplier against a leakage-optimistic

ceiling; it survives even the most favorable trivial baseline this paper could construct.

TimesFM-3’s short-lead CRPS advantage reverses within the tested lead-time range – first sign flip at 22.36 h, durable reversal by 43.49 h (Section 4.3) – and the mechanism behind that short-lead advantage has two independent parts, both now tested separately. The observation-history part does not support an information-asymmetry explanation: a persistence-augmented EMOS at 0–24 h closes only 3.4% of the gap toward TimesFM-3. This result is not treated as confirming an architectural explanation instead, since the test itself has three limitations that keep it from cleanly distinguishing the two hypotheses – a lead-pooled EMOS fit, a single linear persistence term against TimesFM-3’s much richer context window, and a metric-dependent result (EMOS is significantly closer to nominal coverage in the same bucket where TimesFM-3 wins on CRPS) (Section 5.3). The ensemble-spread/calibration part has also now been tested directly: TimesFM-3’s covariate channel does accept a second, simultaneous input, and the model is genuinely sensitive to that channel’s real informational content, not merely its presence (Section 3.2, Section 5.7) – but supplying ensemble spread this way does not cleanly close the calibration deficit. CRPS worsens from 1.3592 K to 1.3837 K; coverage moves closer to nominal, from 0.7603 to 0.8027 – a genuine calibration shift, not merely a wider interval – but interval width also increases substantially, from 5.1379 K to 5.8246 K, the expected cost of that shift under this paper’s own standard of judging sharpness jointly with calibration, not independent evidence of a worse model (Section 5.4). The net effect is a shift along the calibration-sharpness trade-off together with a worse CRPS, not a clean improvement or a clean failure. Neither half of the mechanism therefore supports the information-asymmetry framing – but neither confirms architecture as the explanation either: the channel accepted the second input and was demonstrably sensitive to it, and did shift its behavior in response, yet that shift did not convert into a net improvement within this study’s zero-shot, frozen-weights setup. The short-lead advantage’s mechanism remains an open interpretive question, with architecture one candidate explanation among others, not a confirmed one.

Two questions that were open earlier in this project are now both resolved. EMOS’s training/test ensemble-size mismatch is bounded and too small to be a material contributor to the calibration-sharpness comparison (Section 5.7). And whether ensemble spread could be supplied to TimesFM-3 as a second covariate, and whether doing so would change its calibration, has been tested directly (Section 5.4): it can be supplied, but doing so does not improve calibration in a way that supports the information-asymmetry framing (Section 5.3). Neither resolution changes this paper’s two central findings: on CRPS, a frozen, zero-shot time series foundation model is not competitive with a few days of station-specific training data, for either a trained statistical postprocessing method or a trivial one-parameter rescaling of the raw ensemble (though the exact training size at which this becomes statistically significant, rather than a raw point-estimate crossing, depends on block definition, Section 4.5); and on calibration-sharpness, TimesFM-3’s advantage over even a leakage-optimistic trivial rescaling is real. The evidence for both is for one time series foundation model, TimesFM-3; whether it generalizes to other pretrained time series models (e.g. Chronos-2, Moirai-2) is untested. Both findings are also specific to the target variable studied here, 2m temperature, where EMOS’s Gaussian predictive family is close to ideally specified; for targets requiring non-Gaussian or target-specific reformulations, a trained method’s small-sample advantage should not be assumed to appear as early.

Code and Data Availability

Code for this study is publicly available at github.com/elendmire/zero-shot-postprocessing. A versioned, archival release of this code is available via Zenodo (concept DOI: 10.5281/zenodo.22662543, which always resolves to the latest version; verified by direct Zenodo record inspection, 2026-09-08). EUPPBench, version v1.0, is described in Demaeyer et al. [2023] and distributed via Zenodo (DOI: 10.5281/zenodo.7429236, verified by direct DOI resolution and

Zenodo record inspection, 2026-09-07); this study uses its own defined train/test split without modification (Section 2). TimesFM 3.0 weights (checkpoint `google/timesfm-3.0-pytorch`) are obtained from the model card at <https://huggingface.co/google/timesfm-3.0-pytorch> (Section 3.2), distributed under the TimesFM Non-Commercial License v1.0 (verified against the model card’s own license field, 2026-09-07); this study’s use is academic and non-commercial, consistent with that license.

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CRedit Author Contributions

Ömer Faruk Avcı: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. Sole author; all roles.

Competing Interests

The author declares no competing interests.

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